



# Hadoop Cluster Notes & References

*Key commands, AI assistance usage & important resources*

|                |                                   |
|----------------|-----------------------------------|
| Sections       | 3 — Commands, AI Usage, Resources |
| Hadoop Version | Hadoop 3.4.2                      |
| ZooKeeper      | ZooKeeper 3.8.6                   |
| Java Runtime   | OpenJDK 17                        |
| Environment    | Docker Containers                 |

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## 1. What I Used AI For

AI assistance was used selectively throughout this project — not as a replacement for understanding, but as a productivity tool for specific tasks.

1

### Formatting Documents & PDFs

Used AI to polish documentation with consistent structure, correct grammar mistakes, and produce professionally formatted Word and PDF files — including this document.

2

### Environment Files

AI helped generate clean, well-structured environment variable files (.bashrc, /etc/profile.d/ scripts, and Docker env\_file configs) with correct syntax and ordering.

3

### Automating Recovery After Container Failures

After losing the cluster setup 3 times due to docker-compose down, AI helped design the docker-compose.yml structure and startup scripts to automate service initialization on container restart.

4

### Understanding Errors & Log Analysis

When encountering cryptic errors (like "Connection refused" during NameNode format, or SIGHUP killing daemons), AI helped interpret log output and pinpoint the root cause quickly.

5

### Writing perfect README file on GitHub

AI was used to write clear, accurate README md file on GitHub with all setup guidance.

## 2. Most Important Commands

These are the essential commands used throughout the cluster setup, monitoring, and maintenance process. Each command is paired with an explanation of why it matters.



### Log Inspection — HDFS & YARN Daemons

```
$ cat /opt/hadoop/logs/*namenode*.log | tail -50
```

- Shows the last 50 lines of the NameNode log. Critical for diagnosing NameNode startup failures, QJM connection issues, and failover events.

```
$ cat /opt/hadoop/logs/*datanode*.log | tail -50
```

- Inspects DataNode logs for block-reporting errors, connection timeouts to the NameNode, or disk failures.

```
$ cat /opt/hadoop/logs/*journalnode*.log | tail -50
```

- ▶ JournalNode is the backbone of HA edit-log synchronization. Checking this log is the first step when NameNode format or startup fails.

```
$ cat /opt/hadoop/logs/*zkfc*.log | tail -50
```

- ▶ ZKFC manages automatic failover via ZooKeeper. This log reveals health-check failures and fencing decisions between Active and Standby NameNodes.

```
$ cat /opt/hadoop/logs/*resourcemanager*.log | tail -50
```

- ▶ The ResourceManager log tracks job scheduling, NodeManager registrations, and YARN HA failover events.

```
$ cat /opt/hadoop/logs/*nodemanager*.log | tail -50
```

- ▶ NodeManager handles container launches on worker nodes. Useful for debugging failed MapReduce tasks or resource allocation errors.

```
$ ls /opt/hadoop/logs/
```

- ▶ Lists all log files present. Useful to confirm which daemons are running and find the exact log filename before tailing.



## Docker Image Management

```
$ docker images | findstr hadoop
```

- ▶ Lists all locally saved Hadoop images. Essential for verifying that docker commit snapshots were created successfully before running docker-compose down.

```
$ docker commit node01 hadoop-node01:latest
```

- ▶ Saves the full state of node01 (all installed packages, configs, and binaries) as a new Docker image. This is the key solution to surviving docker-compose down without losing the entire Hadoop setup.

```
$ docker commit node02 hadoop-node02:latest
```

- ▶ Saves the state of node02 — the Standby NameNode and second ZooKeeper participant. Must be committed separately since each node has a unique myid for ZooKeeper.

```
$ docker commit node03 hadoop-node03:latest
```

- ▶ Saves node03 which hosts the third ZooKeeper instance and the third JournalNode alongside its DataNode/NodeManager roles.

```
$ docker commit node04 hadoop-worker:latest
```

- Saves the worker node image (node04 and node05 share the same worker role — no ZooKeeper or JournalNode, just DataNode + NodeManager).

## Cluster Health Checks

```
$ hdfs haadmin -getAllServiceState
```

- Reports whether each NameNode is Active or Standby. The single most important command to verify HDFS HA is working correctly and that automatic failover is functioning.

```
$ yarn rmadmin -getAllServiceState
```

- Same as above but for YARN ResourceManagers. Confirms that one RM is Active and the other is in Standby, ensuring job scheduling continues if one RM fails.

```
$ jps
```

- Java Process Status — lists all running JVM processes on the current node. A quick sanity check to see which Hadoop/ZooKeeper daemons are actually alive (NameNode, DataNode, JournalNode, QuorumPeerMain, etc.).

## 3. Important Resources

The following references were essential during the research, setup, and troubleshooting phases of this project.

1

### Apache Hadoop 3.4.2 — HDFS High Availability Using the Quorum Journal Manager

<https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-hdfs/HDFSHighAvailabilityWithQJM.html>

2

### Apache Hadoop 3.4.2 — HDFS Users Guide

<https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-hdfs/HdfsUserGuide.html>

3

### Apache Hadoop 3.3.6 Installation on Ubuntu 22.04 — Abhik Dey, Medium

<https://medium.com/@abhikdey06/apache-hadoop-3-3-6-installation-on-ubuntu-22-04-14516bceec85>

4

### How to Set Up Apache Hadoop and YARN: A Step-by-Step Guide — Towards Dev

<https://towardsdev.com/how-to-set-up-apache-hadoop-and-yarn-a-step-by-step-guide-8e939f1a4e44>

5

### Apache Hadoop Community Wiki (Confluence)

|                                     |                                                                                                                       |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Hadoop Cluster — Notes & References | Technical Reference                                                                                                   |
|                                     | <a href="https://cwiki.apache.org/confluence/display/hadoop/">https://cwiki.apache.org/confluence/display/hadoop/</a> |